

$$700 \times 10^{-9} \text{ m}$$

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Ans: B

A: Since Y and Z is the same distance from charge Q, the potential at both point is the same. Having no potential difference, no work is done to taking a charge from Y to Z.

B: Electric field radiate outwards from the positive charge, hence the field strength acts in a direction from X to Z.

C: Since Y and Z is the same distance from charge Q, the magnitude of the electric field strength at both points is the same.

D: Since Y and Z is the same distance from charge Q, the potential at both point is the same.

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Ans: B